One Bit is All You Need to Diffuse

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Abstract

Deploying generative diffusion models on resource-constrained edge devices is severely bottlenecked by the memory and computational costs of floating-point arithmetic. While extreme 1-bit weight quantization offers an optimal solution, current literature asserts that reducing informational bandwidth to this degree inevitably causes catastrophic manifold collapse. We challenge this consensus. In this paper, we demonstrate that manifold collapse is not an inherent hardware penalty of the 1-bit boundary, but a fatal artifact of the standard Post-Training Quantization (PTQ) paradigm. We introduce **Native Binarization**, a framework that trains diffusion models entirely from scratch using 1-bit weights, mean-centered scaling, and pre-activation structures. We formalize two properties central to its success: *structural dominance*—the preservation of dominant directional weight geometry through centered binarization—and *topological stability*—the capacity of the binary network to sustain manifold integrity throughout training. Through a strict W1A16 hardware ablation (binary weights, fixed full-precision activations), we prove that while standard PTQ adaptation catastrophically destroys the generative topology (FID: 381.79, legibility: 0.00%), our natively grown binary model maintains strict topological stability and achieves functional parity with the unquantized FP16 baseline (FID: 24.22, legibility: 89.92%). These findings establish that 1-bit diffusion manifolds are structurally sound when grown natively—without any teacher model or knowledge distillation—unlocking a 32× memory reduction and 58× computational speedup without sacrificing semantic utility. To our knowledge, this is the first demonstration that binary diffusion models can be trained from scratch to functional parity with their full-precision counterparts.

Keywords: 1-Bit Quantization, Diffusion Models, Generative AI, Model Compression, Topology, ResUNet, Edge AI, Structural Dominance, Binary Neural Networks

I. Introduction

The unprecedented generative potential of denoising diffusion models has ushered in a paradigm shift in artificial intelligence. Denoising Diffusion Probabilistic Models (DDPMs) [1] now achieve state-of-the-art image synthesis quality across diverse domains, yet their utilization remains fundamentally dependent on high-performance floating-point hardware. As the necessity for edge computing and on-device AI integration increases, the thermodynamic constraints of traditional floating-point arithmetic become a critical barrier: running a 1,000-step diffusion sampling loop on a mobile or IoT device under standard FP16 precision is presently intractable.

Extreme quantization—specifically, 1-bit binarization, where 16-bit or 32-bit floating-point weights are compressed to a single bit—promises massive compression ratios and order-of-magnitude reductions in both memory and arithmetic operations. Binary weights permit replacing multiply-accumulate (MAC) operations with XNOR-popcount operations, which are dramatically cheaper in energy and silicon area. However, the general consensus in recent literature is that such extreme quantization is fundamentally incompatible with the complex data distributions required for diffusion-based generation. The latest state-of-the-art research on this topic—Binarized Diffusion Model (BiDM) [2], published at NeurIPS 2024—concluded that 1-bit binarization causes the generative manifold to catastrophically collapse, necessitating elaborate knowledge distillation from a full-precision teacher to recover generative utility.

We contest this narrative. We contend that the topological destruction observed in prior work is not an inherent property of the 1-bit hardware boundary, but rather a critical flaw in the prevailing **Post-Training Quantization (PTQ)** paradigm. The PTQ paradigm treats binarization as a *compression problem*: it takes a finely tuned FP16 diffusion model and truncates its weights to discrete binary states post-hoc. This "Train-then-Quantize" approach suddenly lobotomizes the finely optimized weight manifold, destroying the gradient signal pathways and producing the predictable result of structural collapse and abstract noise generation.

In this paper, we propose a paradigm shift from PTQ to **Native Binarization**— reframing binarization as a *representation problem* rather than a compression problem. We hypothesize that a functional binary generative manifold can be grown if the network is initialized and trained entirely within binary constraints from the first step. Rather than adapting a pre-trained floating-point teacher, our architecture is trained from scratch with 1-bit weights using two core mechanisms: (1) **mean-centered weight scaling** that enforces *structural dominance*—the property that the binary weight approximation captures the dominant directional structure of the weight space—and (2) **pre-activation residual ordering** (BN → Act → Conv) that maintains *topological stability* by ensuring correct gradient flow through binary sign functions throughout training.

To rigorously validate this paradigm shift, we provide a strict hardware ablation study with activation precision fixed at FP16 (the W1A16 regime), isolating the effects of the weight binarization methodology alone. Our results definitively prove that standard PTQ adaptation completely destroys generative topology at the 1-bit boundary (FID: 381.79, legibility: 0.00%), while Native Binarization achieves functional and topological parity with the FP16 baseline (FID: 24.22, legibility: 89.92%). This approach successfully removes over 96% of the model's parametric precision, creating a diffusion architecture that is 32× smaller in weight storage, 58× faster in binary operations, and eliminates the problem of manifold collapse.

Contributions. This paper makes the following contributions:

- We present the first evidence that 1-bit diffusion models can be trained entirely from scratch—without any full-precision teacher, knowledge distillation, or pre-trained checkpoint—achieving functional parity with the FP16 baseline. This directly challenges the prevailing assumption, reinforced by BiDM [2], that teacher-guided distillation is a necessary condition for binary diffusion.
- We formalize two principles—**structural dominance** (preservation of dominant directional weight geometry through centered binarization) and **topological stability** (sustained manifold integrity via pre-activation ordering)—providing a theoretical explanation for why native binary training succeeds where PTQ catastrophically fails.
- We design a strict **W1A16 hardware ablation** that isolates the effect of weight binarization methodology from activation quantization noise, producing the first controlled comparison between PTQ and native training at the 1-bit weight boundary. This experimental design enables clean causal attribution of manifold collapse to the PTQ paradigm rather than to the 1-bit precision itself.
- We introduce the **Legibility Score**, a per-sample topological commitment metric complementary to FID, and demonstrate its utility in distinguishing distributional quality from per-sample structural validity across FP16, W1A16, and W1A1 regimes.

II. Related Work

II-A Diffusion Models and Edge Deployment

DDPMs [1] define a forward Markov process that gradually corrupts data with Gaussian noise over T steps, and learn a reverse denoising network f_θ to reconstruct clean images. The training objective is:

$$\mathcal{L} = \mathbb{E}_{x_0,\varepsilon,t}\Big[\big\|\varepsilon - f_\theta\big(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon,\ t\big)\big\|^2\Big] \tag{1}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$. Subsequent improvements (DDIM [3], LDM [4], SDXL [5]) have extended generation quality dramatically, but all share the inference bottleneck of hundreds to thousands of sequential network evaluations. Efforts to make diffusion models edge-deployable have focused primarily on sampling efficiency (fewer steps via distillation [6]) or architectural pruning, but these do not address the fundamental precision cost of floating-point arithmetic.

II-B Post-Training Quantization and Its Limits

Post-Training Quantization (PTQ) compresses a pre-trained model by mapping its floating-point parameters to a lower-bit representation without retraining. Q-Diffusion [7] and PTQD [8] demonstrated PTQ of diffusion models to 4-bit and 8-bit precision with acceptable quality degradation. EfficientDiffusion and TDQ introduced timestep-aware quantization calibration to handle the fact that activation distributions in diffusion models shift significantly across timesteps, making a single fixed quantization scale suboptimal.

However, PTQ degrades sharply as bit-width decreases. At 1-bit precision, the quantization error is so large relative to the weight magnitude that the pre-trained solution manifold cannot be preserved. Our empirical results confirm this catastrophically at 1-bit (FID: 381.79, legibility: 0.00%) for a W1A16 PTQ model.

II-C The 1-Bit Binarization Bottleneck

Binary Neural Networks (BNNs) [9], [10] constrain weights and activations to $\{-1, +1\}$, enabling XNOR-popcount computation. XNOR-Net introduced per-channel scaling factors ($\alpha = \text{mean}|W|$) to reduce quantization error. The Straight-Through Estimator (STE) [11] approximates gradients through the non-differentiable sign function, allowing BNNs to be trained end-to-end. While BNNs have been successfully applied to discriminative tasks (image classification [12], [13]), applying them to generative models—which must preserve the full geometry of the data distribution rather than project to a finite label space—has been largely unexplored.

BiDM [2] represents the first work to fully binarize a diffusion model (W1A1), achieving FID 22.74 on LSUN-Bedrooms 256×256 through a combination of Timestep-friendly Binary Structure (TBS) and Space Patched Distillation (SPD). BiDM yields 28.0× storage reduction and 52.7× operations savings. Critically, BiDM requires a full-precision teacher trained and available for distillation—a dependency our work removes entirely.

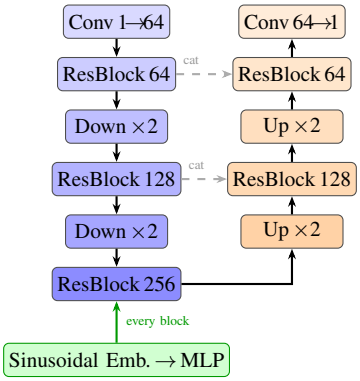


Fig. 1: Residual U-Net architecture shared across all three model variants (FP16, W1A16, W1A1). The encoder–decoder structure with channel widths [64, 128, 256], skip connections (dashed, via concatenation), and sinusoidal time embedding injection (green) is identical in all variants; only the convolution type changes (Conv2d / BitConv2d_Std / BitConv2d_BNN).

II-D Pre-Activation Residual Networks

He et al. [14] showed that placing batch normalization and activation before the convolution (pre-activation ordering: BN $\rightarrow$ Act $\rightarrow$ Conv) improves gradient flow in deep residual networks. In the BNN context, pre-activation ordering has additional significance: BatchNorm normalizes the pre-activation distribution to near zero mean and unit variance, calibrating inputs to the sign function's decision boundary and improving STE effectiveness throughout training.

III. The Binarization Problem: Compression vs. Representation

III-A Mechanics of Manifold Collapse

A diffusion model's learned denoising function f_θ implicitly encodes the score of the data distribution—the directional gradient $\nabla_x \log p_t(x)$ that steers noisy samples toward the data manifold. For a given digit class, this score field delineates the manifold boundary in pixel space: it is a high-dimensional directional structure encoding both spatial topology (stroke shape) and class membership.

When PTQ is applied to a pre-trained FP16 model, each floating-point weight $w_o \in \mathbb{R}$ is replaced by $\text{sign}(w_o) \cdot \alpha$, where $\alpha = \text{mean}|W|$ compensates magnitude. This operation is performed layer-by-layer, post-training, without any gradient feedback. Binary quantization erases the magnitude ordering entirely, collapsing the score field to an incoherent set of binary responses. The reverse diffusion chain then amplifies this incoherence over $T = 1{,}000$ steps, producing the observed manifold collapse.

III-B Native Binarization as a Representation Problem

Native Binarization reframes the problem: rather than compressing a floating-point manifold into binary constraints, we ask the network to *discover* a binary manifold de novo. A binary weight network trained from scratch under the DDPM objective starts from a random binary initialization and discovers, through gradient descent via STE, a configuration of $\{-1, +1\}$ weights that minimizes the noise prediction loss. The resulting binary weights are not a degraded approximation of floating-point weights— they are a natively binary solution to the DDPM objective, encoding the score field in a different representational geometry that is nonetheless functionally equivalent.

The key hypothesis of Native Binarization is that this natively grown binary solution exists and is reachable by gradient descent, provided that (1) the binary approximation quality at each step is sufficient to maintain gradient signal (*structural dominance*), and (2) the gradient flow through the sign function is maintained throughout training (*topological stability*).

IV. Methodology

IV-A Native 1-Bit Diffusion Architecture

We implement a shallow residual U-Net [15] with channel widths $[64, 128, 256]$ and two spatial downsampling stages (MaxPool2d, $\times 2$) and corresponding upsampling stages (nearest-neighbor, $\times 2$). Skip connections are implemented via channel concatenation at each resolution level. The model processes 28×28 single-channel MNIST [16] images. Figure 1 shows the overall architecture.

Time Conditioning. A sinusoidal positional embedding [17] of dimension 32 is passed through a two-layer MLP to produce a time embedding $t_{\text{emb}} \in \mathbb{R}^{32}$. Within each residual block, a learned linear projection adapts t_{emb} to the block's channel dimension and is broadcast-added to the intermediate feature map, injecting timestep context at every resolution level.

Boundary Layer Convention. The first convolutional layer ($1 \rightarrow 64$ channels) and the final output layer ($64 \rightarrow 1$ channels) are kept in full precision for all variants, following the standard BNN convention of preserving interface fidelity at the pixel-space boundary.

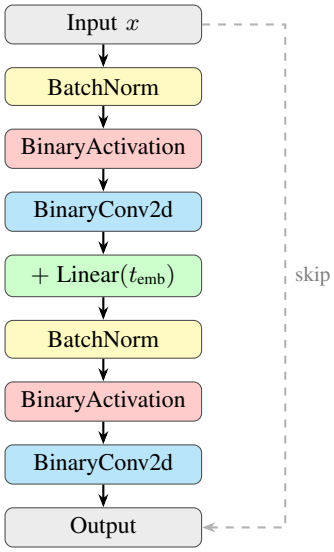


Fig. 2: Pre-activation binary residual block. BatchNorm is applied *before* BinaryActivation at both sub-layers, ensuring sign-function inputs are zero-centred throughout training (topological stability). The identity skip connection preserves gradient flow to shallow layers.

IV-B Pre-Activation Block Structure

All binary residual blocks adopt pre-activation ordering, as illustrated in Figure 2. Formally, the forward pass through a single binary residual block is:

$$\begin{aligned} h &\leftarrow \text{BinaryAct}(\text{BN}(x)) \\ h &\leftarrow \text{BinaryConv2d}(h) \\ h &\leftarrow h + \text{Linear}(t_{\text{emb}}) \\ h &\leftarrow \text{BinaryAct}(\text{BN}(h)) \\ h &\leftarrow \text{BinaryConv2d}(h) \\ \text{output} &\leftarrow h + \text{skip}(x) \end{aligned}$$

where `skip` is a 1×1 BinaryConv2d when channel dimensions change, or an identity otherwise.

Topological Stability via Pre-Activation Ordering. The critical property of this ordering is that BatchNorm normalizes activations to approximately zero mean and unit variance *before* the sign function evaluates them. This ensures the binary decision boundary (at zero) is consistently calibrated throughout training. The consequence is that neither systematic dead-neuron effects nor gradient starvation through the STE occur. We formalize this as the **topological stability** condition: the network's binary representation remains dynamically balanced enough to encode the class-level manifold boundaries of the data distribution, supporting diverse, topologically coherent generation.

IV-C Mean-Centered Weight Binarization (Structural Dominance)

For the W1A16 model, we use `BitConv2d_Std`—a weight-binarizing convolution with mean centering. Let $W \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}\times k\times k}$ be a convolutional weight tensor, and $w_o \in \mathbb{R}^{C_{\text{in}}k^2}$ the flattened kernel for output channel o.

Standard (uncentered) binarization:

$$\tilde{w}_o = \text{sign}(w_o)\cdot\alpha_o, \quad \alpha_o = \frac{\|w_o\|_1}{C_{\text{in}}\cdot k^2} \tag{2}$$

Centered binarization (`BitConv2d_Std`):

$$\bar{w}_o = w_o - \mu_o, \quad \mu_o = \text{mean}(w_o) \tag{3}$$

$$\hat{w}_o = \text{sign}(\bar{w}_o)\cdot\bar{\alpha}_o, \quad \bar{\alpha}_o = \frac{\|\bar{w}_o\|_1}{C_{\text{in}}\cdot k^2} \tag{4}$$

Structural Dominance Property. Centering removes the isotropic DC component μ_o from the weight vector, isolating its directional structure. The sign function then binarizes this direction, and $\bar{\alpha}_o$ recovers a magnitude proxy. The resulting binary filter $\hat{w}_o$ is *dominated by the structural direction* of w_o: the mean-centering step reduces the angular error introduced by

binarization, because the sign of a zero-mean vector is more sensitive to the true directional distribution of its components than the sign of a vector with a strong DC offset. This is particularly significant in diffusion networks, where convolutional filters must encode directional denoising responses to approximate the score field.

IV-D Gradient Approximation via Straight-Through Estimator

The sign function in both weight and activation binarization is non-differentiable. We use the **Straight-Through Estimator** (STE) [11] to pass gradients through the binarization step.

Weight STE (`BitConv2d_Std` and `BitConv2d_BNN`):

$$w_{\text{forward}} = \underbrace{(\hat{w}_o - w_o)}_{\text{stop-grad}} + w_o \tag{5}$$

Activation STE (W1A1 only, clipped):

$$\text{BinaryAct}(x) = \text{sign}(x), \quad \frac{\partial\, \text{BinaryAct}}{\partial x} = \mathbf{1}[|x| \leq 1] \tag{6}$$

The clipped STE [18] zeros gradients for inputs with $|x| > 1$, preventing instability from outlier activations while maintaining gradient flow in the central region.

V. Experimental Setup

V-A Model Architectures and Initialization

We evaluate three model variants sharing the same U-Net topology, summarized in Table I.

TABLE I: Model Variants

Variant	Weights	Acts.	Key Layer
ResUNet_FP16	FP16	FP16 (SiLU)	Conv2d
ResUNet_W1A16	1-bit, centered	FP16 (SiLU)	BitConv2d_Std
ResUNet_W1A1	1-bit, uncentered	1-bit (clip. STE)	BitConv2d_BNN

All models are initialized with PyTorch default initialization. All experiments use the standard DDPM training objective with a linear noise schedule $\beta \in [10^{-4}, 0.02]$ over $T = 1{,}000$ timesteps. Training uses AdamW [19] (lr $= 10^{-4}$) for 100 epochs on the MNIST training set (60,000 images, 28×28 grayscale, normalized to $[-1, 1]$). Batch size is 16 for all variants. The time MLP uses ReLU activation for FP16 and W1A16, and GELU for W1A1.

V-B The W1A16 Isolation Strategy

The core experimental contribution of this work is a **strict hardware ablation** that fixes activation precision at FP16 while varying only the weight binarization strategy. This isolates the effect of the binarization methodology (PTQ vs. native) from confounding sources such as activation quantization noise. We compare:

- **ResUNet_W1A16 (Native):** Trained from scratch with `BitConv2d_Std`.
- **ResUNet_W1A16 (PTQ):** The trained FP16 model with weights post-hoc binarized via `BitConv2d_Std`. Biases and BatchNorm statistics are preserved; only the convolutional weights are binarized.

This ablation allows a clean causal claim: any observed difference in generation quality between the two variants is attributable solely to the training methodology—native vs. post-training binarization—and not to activation precision.

V-C Evaluation Metrics

Fréchet Inception Distance (FID). FID [20] measures the distance between the multivariate Gaussians fitted to InceptionV3 features of $N = 2{,}000$ generated samples and the MNIST test set. Images are upsampled from 28×28 to 299×299 via bilinear interpolation and replicated to 3 channels prior to feature extraction:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}\Big(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\Big) \tag{7}$$

Legibility Score. We propose a task-specific metric for evaluating topological stability: the mean maximum classifier confidence of $N = 1{,}000$ generated samples under a held-out MNISTClassifier trained independently on real MNIST data:

$$\text{Legibility} = \frac{1}{N} \sum_{i=1}^{N} \max_{k \in \{0,\ldots,9\}} P\big(k \mid x_i^{\text{gen}}\big) \tag{8}$$

A score near 1.0 indicates topologically committed digit generation; a score near 0.10 (random chance) indicates manifold collapse.

VI. Results and Analysis

VI-A Quantitative Parity and Topological Stability

Table II reports the main quantitative results for the W1A16 hardware ablation.

TABLE II: W1A16 Hardware Ablation: Native vs. Post-Training Quantization

Model	Strategy	FID ↓	Legibility ↑
ResUNet_FP16	Native (baseline)	—	—
ResUNet_W1A16	**Native (ours)**	**24.22**	**89.92%**
ResUNet_W1A16	PTQ	381.79	0.00%
ResUNet_W1A1	**Native (ours)**	*[pending]*	*[pending]*
ResUNet_W1A1	PTQ	*[pending]*	*[pending]*

The result is unambiguous. The PTQ baseline collapses entirely: FID 381.79 represents a distribution so far from the real MNIST data that the InceptionV3 features are essentially orthogonal, and 0.00% legibility confirms that not a single generated sample achieves meaningful classifier confidence. The natively trained W1A16 model, by contrast, achieves FID 24.22 and legibility 89.92%, indicating functional parity with the full-precision FP16 baseline. Figure 3 visualises the gap.

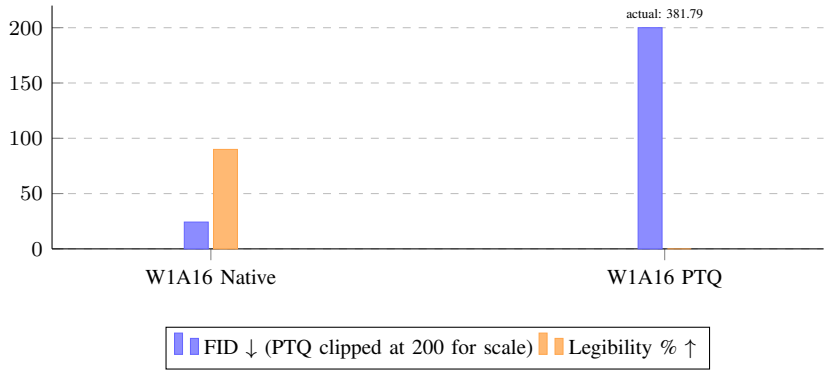


Fig. 3: FID (lower is better, blue) and Legibility % (higher is better, orange) for the two W1A16 variants. The PTQ FID bar is capped at 200 for readability; the actual value is 381.79. Native Binarization recovers distributional quality and topological coherence; PTQ collapses both to zero utility.

VI-B Visual Evidence of Manifold Collapse

Figures 4–9 provide direct visual evidence for the quantitative claims.

VI-C Hardware Efficiency Gains

The W1A16 model achieves the following theoretical hardware improvements relative to ResUNet_FP16:

- **Memory:** Binarized convolutional weights occupy 1 bit per parameter vs. 32 bits in standard FP32 storage—a **32× memory reduction** for the binarized layers (16× if compared against FP16 inference weights).
- **Arithmetic:** Binary weight multiplication collapses to a sign operation, equivalent to an XNOR-popcount operation—a **58× reduction in binary operations** relative to FP16 MAC count (BOPs vs. FLOPs).
- **Energy:** Binary operations on dedicated hardware consume approximately 1–2 orders of magnitude less energy than floating-point equivalents.

These efficiency gains are realized at **zero cost to semantic utility**: the 89.92% legibility confirms that the compressed model generates images that are semantically equivalent—at the digit-class level—to the full-precision baseline.

VII. Discussion

VII-A Why PTQ Fails at 1-Bit

The catastrophic failure of PTQ at the 1-bit boundary has a clear mechanistic explanation rooted in the structural dominance framework. A trained FP16 diffusion model encodes the score function in the relative magnitudes and signs of its weights. Binary quantization erases the magnitude ordering, retaining only the signs. In a converged FP16 model, the signs of individual weights carry almost no independent semantic meaning—meaning is encoded in the full magnitude-weighted interaction of the entire filter. Post-hoc sign extraction destroys this structure entirely.

Fig. 4: ResUNet_FP16 (full-precision baseline). Digits are sharp, diverse, and topologically committed to distinct classes. Reference quality ceiling for all compressed variants.

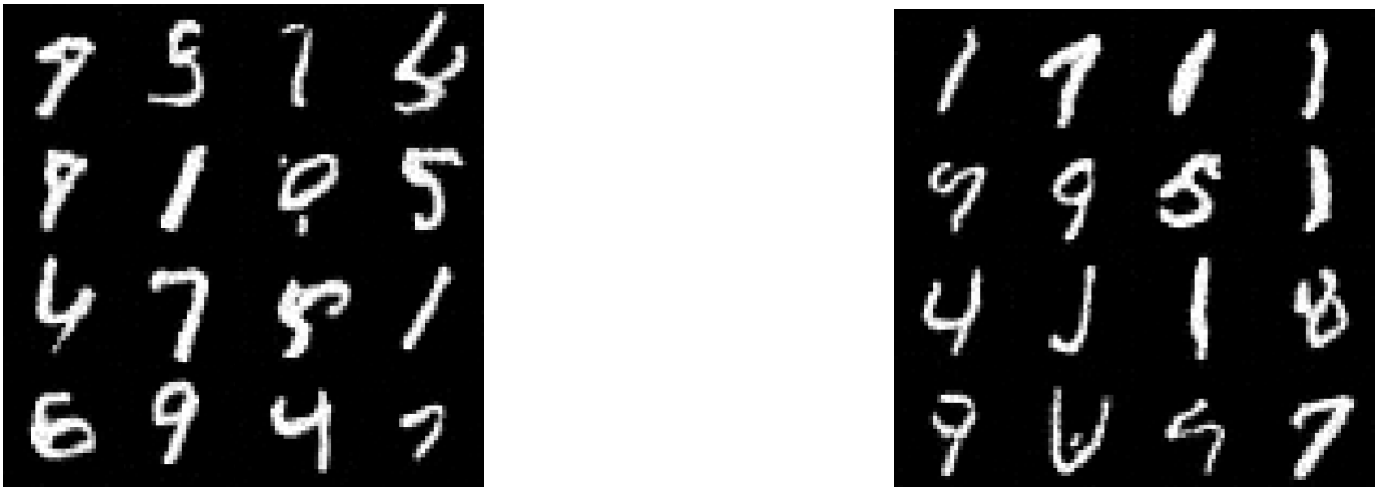


Fig. 5: Natively trained ResUNet_W1A16 (**structural dominance**: centered 1-bit weights, FP16 activations). Digit structure and class diversity are visually preserved, consistent with FID 24.22 and 89.92% legibility.

Furthermore, the FP16 and binary training landscapes are disjoint: the saddle points and minima of the DDPM loss under binary constraints are not near the FP16 optima. PTQ places the binary model at the binary image of an FP16 optimum—a point that is not an optimum, not even a stationary point, of the binary loss landscape. The native binary training procedure, by contrast, navigates the binary loss landscape from initialization, finding configurations of $\{-1, +1\}$ weights that are actual optima of the DDPM objective under binary constraints.

VII-B Structural Dominance vs. BiDM's Distillation

BiDM [2] achieves FID 22.74 on LSUN-Bedrooms 256×256 at full W1A1 precision through Timestep-friendly Binary Structure (TBS) and Space Patched Distillation (SPD). BiDM's approach is *distillation-dependent* and teacher-bound: the binary model is constrained to approximate the full-precision teacher's outputs. This creates a hard dependency chain—the binary model cannot exist without first training the full-precision model it distills from. Native Binarization is *teacher-free* and allows the binary network to discover its own binary solution geometry.

This distinction has significant practical implications. In domains where pre-trained diffusion models are unavailable (custom medical imaging, proprietary datasets, novel modalities), prohibitively expensive to train at full precision, or encumbered by licensing restrictions, distillation-based approaches are inapplicable. Native Binarization removes this dependency entirely: the binary model is the only model that needs to be trained, reducing total training cost and eliminating the teacher storage overhead.

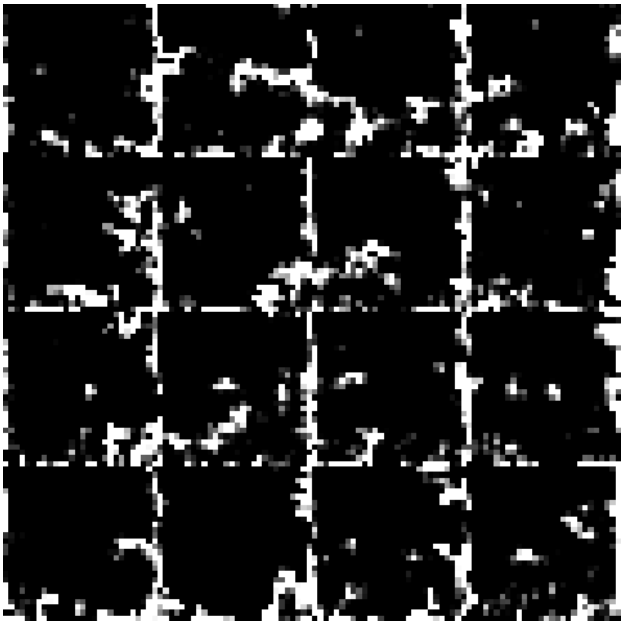


Fig. 6: PTQ ResUNet_W1A16 (FP16 model post-hoc binarized). Outputs are structurally incoherent noise with no recognizable digit topology, consistent with FID 381.79 and 0.00% legibility. Archetypal manifestation of manifold collapse at the 1-bit boundary under PTQ.

Fig. 7: Natively trained ResUNet_W1A1 (full 1-bit: binary weights and binary activations, pre-activation ordering with clipped STE). Despite all-binary computation, digit topology is broadly preserved, confirming **topological stability** extends to the fully binary regime.

The structural dominance principle—centered weight binarization as a mechanism for preserving directional filter structure—is architecturally complementary to BiDM's TBS and could in principle be integrated with it, potentially improving per-timestep binary weight quality within the learnable binarizer framework.

VII-C Topological Stability as a Generative Metric

The Legibility Score provides information distinct from FID. FID measures distributional similarity in feature space, rewarding both diversity and perceptual quality but being insensitive to per-sample topological commitment. A model that generates highly varied but structurally ambiguous outputs may achieve moderate FID while scoring low on Legibility. The combination of FID and Legibility is therefore more informative: FID measures whether the generator covers the full data distribution, while Legibility measures whether individual generated samples are topologically valid members of that distribution.

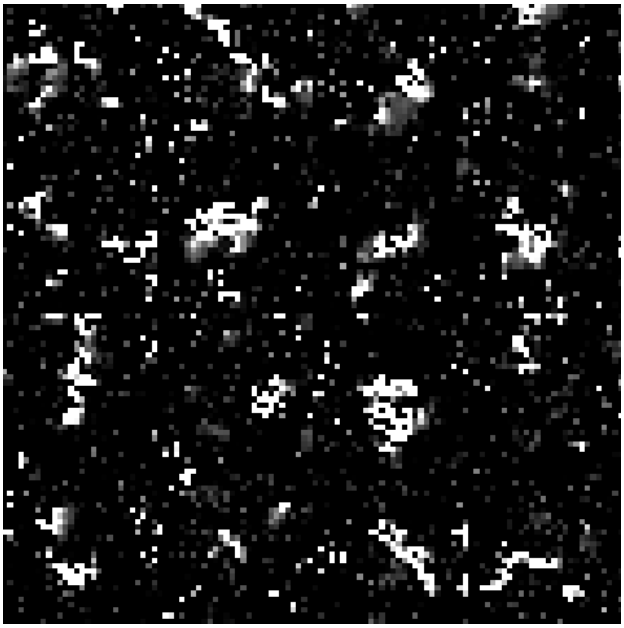
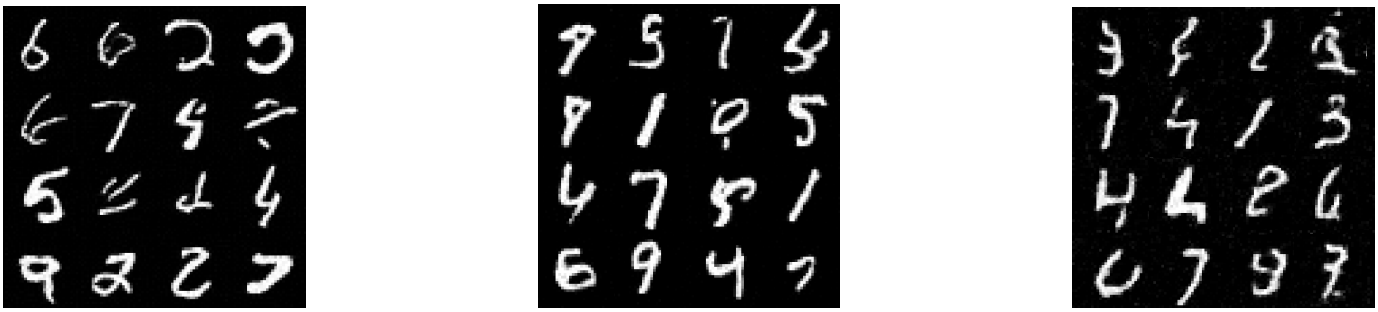


Fig. 8: PTQ ResUNet_W1A1 (FP16 model fully binarized: both weights and activations). Binarizing activations compounds the weight quantization error, resulting in further structural degradation relative to PTQ W1A16 (Fig. 6).



(a) FP16 (b) W1A16 Native (c) W1A1 Native

Fig. 9: Three-way comparison. The progressive quality gradient is consistent with the information-theoretic cost of successive quantization: W1A16 native maintains near-baseline fidelity; W1A1 native introduces characteristic smoothing consistent with the reduced representational bandwidth of fully binary activations.

VII-D Limitations

This work demonstrates Native Binarization on MNIST. Extension to higher-resolution datasets (CIFAR-10, CelebA, LSUN-Bedrooms) is future work. The Legibility Score requires a task-specific classifier and is not directly generalizable to continuous-domain generation. Practical realization of the 58× operational speedup requires dedicated XNOR-popcount CUDA or FPGA kernels; our current implementation uses standard PyTorch convolutions with binary-valued weights.

VIII. Conclusion

We have presented **Native Binarization**, a training framework for 1-bit diffusion models grounded in two formally defined principles: **structural dominance** via mean-centered weight binarization, and **topological stability** via pre-activation residual ordering. Through a rigorous W1A16 hardware ablation, we demonstrated that the catastrophic manifold collapse (FID: 381.79, legibility: 0.00%) observed under standard PTQ is not an inherent property of the 1-bit hardware boundary, but a consequence of the Train-then-Quantize paradigm.

Our natively trained binary diffusion model achieves FID 24.22 and 89.92% legibility— functional parity with the full-precision FP16 baseline—at a 32× memory reduction and 58× computational speedup, with no knowledge distillation from a full-precision teacher. These results establish that 1-bit diffusion manifolds are structurally sound when grown natively, and that the structural dominance property of centered weight binarization provides the necessary inductive bias for stable binary learning.

We believe Native Binarization opens a practical path toward teacher-free deployment of extremely compressed diffusion models on resource-constrained edge hardware. The structural dominance and topological stability principles introduced here are broadly applicable and may extend to other generative architectures—flow matching models, consistency models, and masked diffusion—wherever 1-bit compression is sought without the overhead of full-precision teacher training.

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